



VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN

(Course Code: ECE 1003)

DIGITAL LOGIC DESIGN

Course Title: Digital Logic Design

TPC	3	2	4
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T - Tutorial

P - Practical

C - credit

Mode of Evaluation:

Continuous Assessment Tests - 60%, Practical Assessment - 40%

Continuous Assessment Test-1 20%

Continuous Assessment Test-2 20%

Final Assessment Test-3 20%

Practical Assessment 40%

SYLLABUS CONTENTS

Course Objectives:

1. To learn various techniques for logic minimization.
2. To comprehend the concepts of various combinational circuits.
3. To understand the concepts of various sequential circuits.
4. Helps student to carry out projects which contain digital electronics modules.

SYLLABUS CONTENTS

Course Outcomes:

On completion of the course, students have the ability to,

1. Understand and apply the basic number systems and Boolean algebra in digital design for real world problems.
2. Design and analyze combinational and sequential logic digital circuits.
3. Design and analyze finite state machines.
4. Design and implement Combinational and Sequential circuits using programmable logic devices.
5. Apply this digital logic design knowledge in interdisciplinary projects

SYLLABUS CONTENTS

Module - 1 (**Number Systems and Boolean Algebra**) - 12 Hrs

- ✓ Number systems and conversions, r 's and $(r-1)$'s complement, binary signed and unsigned numbers, binary arithmetic operations
- ✓ Weighted and non-weighted binary codes.
- ✓ Boolean algebra, SOP and POS minimization
- ✓ logic gates and universal gates, tri-state logic and don't care logic, K-maps up to 4-variables.

SYLLABUS CONTENTS

Module - 2 (**Combinational Logic Circuit Design**) - 12 Hrs

- ✓ Binary adder and subtractor, Ripple carry adders/subtractors and fast adders
- ✓ Binary decoders, encoders, multiplexers and de-multiplexers
- ✓ Logic functions using decoders and multiplexers
- ✓ Code converters
- ✓ Magnitude comparator

SYLLABUS CONTENTS

Module – 3 (Sequential Logic Circuit Design) - 13 Hrs

- ✓ Latches, Flip flops, Flip flop conversions
- ✓ Shift Registers
- ✓ counters-synchronous and asynchronous counters, ring and Johnson counters
- ✓ Finite State Machine – design using mealy and Moore state machines
- ✓ Sequence detectors and generators design

SYLLABUS CONTENTS

Module - 4 (**Programmable Logic Devices**) - 5 Hrs

- ✓ PROM, PLA, PAL, Comparison between PROM, PLA and PAL
- ✓ Look-Up Tables (LUT) design

Module - 5 (**Digital Integrated Circuits**) - 6 Hrs

- ✓ Introduction to various logic families
- ✓ Specifications – noise margin, propagation delay, fan-in, fan-out
- ✓ Comparison of logic families.
- ✓ Basic combination circuits design using NMOS, PMOS and CMOS logic families.

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. Samir Palnitkar, “**Verilog HDL: A Guide to Digital Design and Synthesis**,” Prentice Hall, 2nd Edition, 2009.
3. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Thank You